

Training AI to Think Like a Botanist: Using Text-Based Trait Profiles for Interpretable Species Identification in Hyperdiverse Plant Communities

A. Scientific/Technical Goal

Background

Tropical forests harbor most of Earth's terrestrial biodiversity and provide critical ecosystem services, including climate regulation, carbon sequestration, and water cycling^{1,2}. Despite their ecological and economic importance, tropical forests face significant threats from deforestation, habitat fragmentation, and climate change^{3,4}. Understanding, monitoring and predicting how tropical ecosystems respond to these challenges fundamentally relies on precise species identification because each species occupies and performs different roles and functions in the ecosystem⁵⁻⁷. However, achieving species-level identification remains a major hurdle⁸. Botanists often rely on vegetative morphological traits (e.g., leaf arrangement, margin type, venation patterns), which, while diagnostic, are often subtle and variable. As a result, field-based identification is time-consuming and reliant on highly specialized expertise that is not widely accessible in day-to-day ecological monitoring or restoration work. Additionally, in hyperdiverse tropical forests, many species are naturally rare and thus remain underrepresented in image-based training datasets, constraining model scalability. There is a clear need for approaches that integrate the structured botanical knowledge used by taxonomists themselves. This proposal represents a fundamental departure from image-centric AI identification methods, instead leveraging the rich taxonomic knowledge encoded in centuries of botanical literature as the primary reasoning foundation, with imagery serving as complementary validation. This knowledge-driven approach offers the potential for far more interpretable, scalable, and generalizable species identification.

Approach

We are currently extracting taxonomic trait information from both existing structured botanical keys and unstructured natural language descriptions found in floras, field guides, and taxonomic monographs. Soon we will chunk and embed the text data into a vector database for similarity-searches in a retrieval-augmented generation component. Outputs from similarity searches are aggregated with those from a local contrastive language-imaging prediction (CLIP) model we have fine-tuned for pure-image based predictions of underrepresented plant species. Together, these predictions are passed to a reasoning model (GPT-5-Pro / GPT-5) with very particular prompting instructions for comparison, knock-out rules, and confidence thresholding before providing a synthesized taxonomic prediction.

Goal

Our goal is to develop this into an accessible, interpretable agentic-AI workflow that empowers researchers, educators, students, and practitioners to accurately identify tropical plant species, significantly enhancing our capacity to monitor, understand, and manage biodiversity in these critical ecosystems. By doing so, we also aim to contribute to advancements in reasoning-centered AI models by grounding model logic in expert botanical knowledge, helping shape transparent, adaptable tools with broader cross-domain relevance. We expect to have an alpha release of this agentic workflow in Spring 2026 and a beta version for internal grant proposal deadline (December 2026) and submit at least one manuscript within 12 months.

B. Estimate of Compute, Storage, and Other Resources

Our agentic-AI workflow for tropical plant species identification requires substantial computational resources across three distinct experimental phases. Each phase directly supports our objective of developing an accessible, interpretable system that combines text-based botanical knowledge with image assistance for accurate species identification.

Computational Architecture & Models

This submission is a Model Access Request. We are requesting access primarily to the OpenAI credits for GPT-5-Pro API access via NAIRR pilot resources. This bleeding-edge reasoning model provides a 400,000 token context window enabling extensive processing of taxonomic descriptions and superior performance on scientific reasoning tasks. Alternatively, we can utilize GPT-5. The remaining components of the agentic workflow will run on our university compute cluster at the Yale Center for Research Computing. This includes 4TB of permanent storage, 16TB of scratch storage, and a variety of GPU and CPU configurations.

Experimental Design

Our experiments involve iterative refinement of species and family predictions through a multi-agent orchestrated system. Each identification request through the system includes an estimated 80,137.5-tokens for input and 4,000-tokens for output:

- Input = 80,137.5-tokens
 - o 5 candidate taxonomy natural language descriptions
 - For each identification request, the top 5 taxonomic predictions are identified, and their natural language descriptions are extracted and concatenated from a RAG component for input. On average, each description per prediction averages 11,250 words.
 - Using a simplified 1-word = 1.25-tokens, that's 70,312.5-tokens for 5 predictions input text.
 - o 5 high-detail plant images
 - Each image is 2048 x 2048px, which is 765-tokens via the OpenAI vision tokenizer. That's 3,825-tokens in input images.
 - o 1 prompt
 - This includes system prompt, CLIP model probabilities, and orchestration logic. Estimate this to be 1,500-tokens.
 - o Follow-Up Refinement
 - An additional 3 API calls per identification, with confidence-based early exit. Average 1,500-tokens per refinement input. That's 4,500-tokens in total input follow-ups.
- Output = 4,000-tokens
 - o This will be a structured taxonomic identification with confidence scores. Estimate each output to be 1,000-tokens, with up to 3 follow-ups. That's a total of 4,000-tokens in outputs.

Resource Usages & Costs

The costs are calculated per the rates at 1M tokens reported <https://openai.com/api/pricing/> and the previous token estimates. The breakdown between GPT-5-Pro and GPT-5 per a single request are detailed in the following table:

Model	Input \$ / 1M tokens	Input \$ / 1 Request	Output \$ / 1M tokens	Output \$ / 1 Request	Total \$ / 1 Request
GPT-5-Pro	\$15.00	\$1.2020	\$120.00	\$0.48	\$1.6820
GPT-5	\$1.25	\$0.10017	\$10.00	\$0.04	\$0.14017

These calculated costs per request are forecasted into a 3-phase approach for development in the table below:

Phase	Purpose	Test Volume	Primary Model (GPT-5-Pro)	Alternative (GPT-5)
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1: Implementation	Establish workflow, validate architecture	2,000 requests	\$3,364	\$280.34
2: Optimization	Refine prompts, tune thresholds	50,000 requests	\$84,100	\$7,008.50
3: Validation	Full taxonomic evaluation	5,000 requests	\$8,410	\$700.85
TOTAL	Complete Development	57,000 requests	\$95,874	\$7,989.69

C. Support Needs

We do not have need for support.

D. Team and Team Preparedness

Our team is fully prepared to complete the project within the 12-months NAIRR pilot period. We bring together expertise in artificial intelligence, ecology, plant taxonomy, and data science, and includes collaborators from the Yale School of the Environment (YSE) and the Department of Computer Science (CS), as well as New York Botanical Garden (NYBG), aligning directly with this mission.

Prof. Liza Comita (YSE) is a tropical plant ecologist and biodiversity scientist. She will provide access to species-verified field data and ensure ecological relevance throughout the project. Prof. Arman Cohan (CS) is an expert in interpretable machine learning and will lead development of AI models that emulate expert botanical reasoning. Gabrielle Kaili-May Liu (CS) is a PhD student with a background in meta-reinforcement learning and application of human metacognition to LLMs will support RAG corpus and text integration. Mitchell Horn (YSE), Senior Research Data Analyst in YSE, will support advanced data curation, annotation workflows, and model implementation. Nohemi Huanca-Nunez (Associate Research Scientist, YSE) contributes expertise in tropical plant ecology and trait analysis, with practical experience in botanical taxonomy and a growing focus on applying AI tools to ecological research. Dr. Luke Browne (Associate Research Scientist, YSE) has expertise in tropical plant trait variation and ecological data science, including experience with machine learning approaches in ecological contexts. Dr. Fabian Michelangeli, Senior Curator at NYBG and Adjunct Faculty in YSE, brings deep expertise in tropical plant identification and taxonomy.

The team routinely uses Yale's and ACCESS HPC systems with extensive experience with distributed computing, large scientific datasets, multimodal ML workflows, and LMM-based pipelines. All foundational components: clip embeddings, curated image dataset and botanical RAG corpus are completed or near to completion, enabling immediate use of NAIRR provided resources. During months 1-3 we will integrate NAIRR LMM access into exiting clip-based workflow. Months 4-8 will focus on iterative prompt refinement, multiagent orchestration and internal evaluation. Month 9-12 will complete large-batch validation, ecological performance assessment. This structure ensures that all project goals are fully achievable within 12-months NAIRR pilot period.

References: (1) Gibson, L., et al. (2011). *Nature*. (2) Dirzo, R. & Raven, P. H. (2003). *Annual Review of Environment and Resources*. (3) Pan, Y., et al. (2011). *Science*. (4) Hansen, M. C., et al. (2013). *Science*. (5) Petchey, O. L. & Gaston, K. J. (2006). *Ecology Letters*. (6) Engelbrecht, B. M. J. & Comita, L. S. 2007. *Nature* (7) Poorter, L., et al. 2008. *Ecology* (8) Wheeler, Q. D. & Valdecasas, A. G. (2005). *Nature*.